

Review

Anode Materials for Lithium-Ion Batteries, from Conventional Materials to High-Entropy Oxides: A Review of Synthesis Methods, Properties and Sustainability Challenges

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Highlights

What are the main findings?

- Synthesis routes strongly influence LIB anode surface properties;
- Sol–gel methods improve Li⁺ diffusion and cycling stability;
- High-entropy oxides enable stable, high-capacity LIB anodes;
- Green synthesis routes support sustainable battery technologies.

What are the implications of the main findings?

- Sustainable synthesis can reduce the environmental footprint of LIBs;
- Interface-controlled nanostructures enable longer battery lifetime;
- Advanced oxide coatings may replace conventional graphite anodes;
- Hybrid synthesis routes support high-performance energy storage systems.

Abstract

Lithium-ion batteries (LIBs) are essential for current technological infrastructure, driving the development of portable electronics, electric vehicles or grid-scale energy storage. The performance and sustainability of LIBs are critically dependent on their anode materials. This comprehensive review analyzes the evolution and characteristics of key anode materials, highlighting the specific properties they confer to the final battery products. Beyond material properties, the synthesis methods employed for these materials, from conventional techniques (such as solid-state reactions, sol–gel, hydrothermal/solvothermal, co-precipitation, etc.) to innovative and greener approaches (like electrospinning and a novel induction furnace-oxidation hybrid method for complex oxides), are a crucial part in the development of sustainable materials. While these methods offer different advantages, the challenges in achieving optimal electrochemical performance, including issues related to material stability, capacity retention and scalability, remain significant for both research and manufacturing industries. Furthermore, a significant focus is placed on strategies for mitigating the environmental impact associated with anode material production, emphasizing the importance of unconventional and sustainable synthesis routes. Ultimately, the sustainable evolution of LIB technology to achieve future energy demands hinges



Academic Editor: Xuanpeng Wang

Received: 19 June 2026

Revised: 20 July 2026

Accepted: 23 July 2026

Published: 1 August 2026

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on overcoming existing limitations. This necessitates integrated research combining advanced material modeling and design, scalable and environmentally conscious synthesis techniques and in-depth electrochemical characterization.

Keywords: new materials; advanced properties; modeling; environmental impact; synthesis methods

1. Introduction

The use of Li-ion batteries in many different industries has made them an integral part of contemporary daily existence. Thus, the expansion of applications ranging from electronics (smartphones and laptops) to electric vehicles leads to the development of a technology that will underpin the future technological infrastructure. Widespread adoption in various sectors of activity highlights the fundamental role of LIB technology in technological processes, which facilitates the global transition to a sustainable economy. The versatility of LIB plays an important role in the continuous improvement of performance, safety and environmental impact [1].

The operation of a Li-ion battery is based on a complex electrochemical process involving a series of key components, including the cathode, the anode, an electrolyte, and a separator [2]. Among these, the anode material plays a crucial role in determining the overall characteristics of the battery. The choice of anode material significantly influences crucial battery parameters, such as energy density, power density, cycle life, charging rate, and overall safety [3]. Therefore, continuous research and development efforts dedicated to optimizing anode materials are crucial to overcome the current limits of Li-ion battery technology and to meet the increasing demands of energy-intensive applications.

Emerging anode materials, such as high-entropy oxides (HEOs), are attracting increasing interest due to their ability to combine the advantages of several conventional materials, offering a promising balance between structural stability, specific capacity, and operational safety. Compositional optimization of HEOs allows tailoring of electrochemical properties and makes HEOs a growing research area with the potential to overcome many limitations of conventional anode materials and pave the way for next-generation lithium-ion batteries [4–6].

The performance and properties of anode materials are closely related to their synthesis methods. A variety of techniques are used to produce anode materials (sol–gel synthesis, hydrothermal/solvothermal methods, spray pyrolysis, chemical vapor deposition (CVD), etc.), each offering distinct advantages and challenges in terms of control over morphology, crystallinity, purity, and scalability.

As the need for improved energy storage devices increases, the importance of reducing the environmental impact of battery production, including anode components, is becoming increasingly evident. A key aspect is the carbon footprint associated with the materials used in the manufacture of anodes, such as graphite or silicon, which leads to increased efforts to identify sustainable alternatives with reduced environmental impact. In response to these challenges, the scientific community and industry are focusing on developing efficient processes for recycling anode materials and replacing critical raw materials with resources that involve less dependence on fossil fuels. Thus, the increasing focus on sustainability is becoming a key factor in shaping the direction of research in the field of anode materials, driving innovation towards greener solutions, less polluting production technologies and effective end-of-life management strategies. All of this aims to reduce the overall ecological footprint of lithium-ion batteries [7].

This study offers a comprehensive review of the most commonly used materials for Li-ion battery anodes throughout history. It highlights the specific properties these materials impart to the final products, the methods used to achieve optimal battery performance, and the broader implications driven by this continuous need for development.

2. Anodic Materials in Li-Ion Batteries

2.1. Li-Ion Batteries Structure

Generally, a Li-ion battery relies on two electrodes: a cathode, which is positive, and an anode, which is negative. These electrodes are immersed in an electrolyte solution rich in lithium ions. A microporous polymer membrane, acting as a separator, physically separates the electrodes while allowing lithium ions to shuttle between them, preventing electron flow directly. While liquid electrolytes remain common, research on polymer, gel, and ceramic alternatives continues for Li-ion battery applications. The primary components of Li-ion batteries are schematically illustrated in Figure 1.

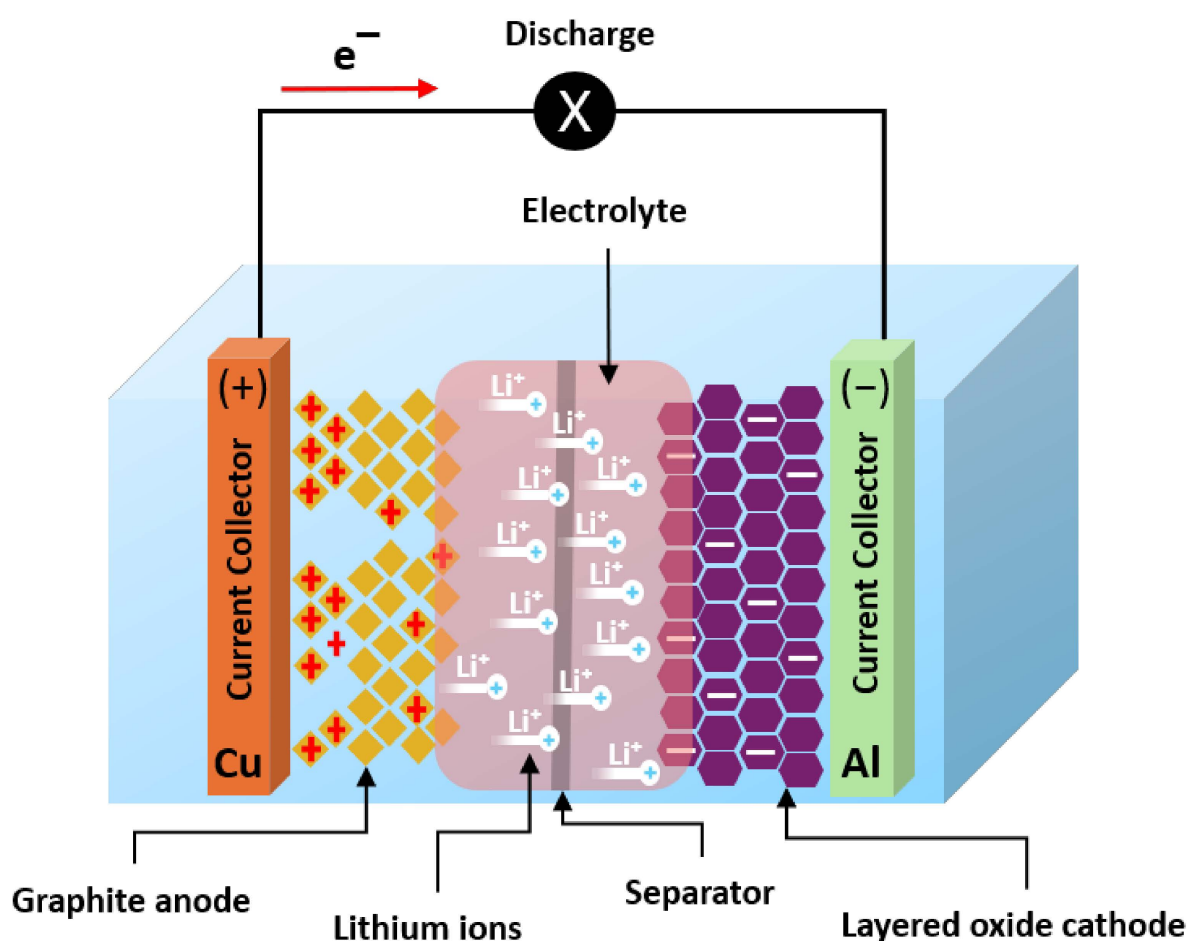


Figure 1. Schematic representation of the main components of a lithium-ion battery during the discharging process: Li^+ ions migrate from the graphite anode to the layered oxide cathode through the electrolyte and separator, while electrons flow from the anode to the cathode through the external circuit (redrawn and adapted from [7]).

Although many studies have been carried out regarding the development of these types of batteries, the working principle has remained the same in the last two decades when it was first developed. On the other hand, the materials' evolution regarding the electrode, electrolytes and separators has brought many advantages in battery development [2].

The anode, one of the most important parts of a lithium-ion battery, plays a key role in the storage and release of energy. Therefore, the process of storing lithium ions and releasing electrons takes place within this crucial component. Basically, there is a lithium ion and electron exchange between the anode and cathode, and this represents the main operation that occurs in the heart of the lithium-ion battery [8].

In recent years, lithium-ion batteries (LIBs) have become a field of interest in research and technology development, being one of the most promising energy storage solutions for the challenges facing present-day society. This ascension was mainly due to the special properties that the anode has, such as high capacity [9,10], stability [11], conductivity [12,13], safety or environmentally friendly properties [14,15].

2.2. Anodic Materials

The anode of a Li-ion battery is usually made from multiple materials that are chosen based on their capacity to store and discharge lithium particles throughout charging and discharging cycles.

2.2.1. Graphite

Lithium has been utilized as a pure metal in crafting anodes for Li-ion batteries since the 1970s. This material demonstrated its suitability for energy storage due to a significant characteristic—its high energy density, with lithium metal achieving an extraordinary theoretical specific capacity of 3861 mAhg^{-1} and 2062 mAhcm^{-3} [16]. However, investigations went beyond energy density, being concentrated on enhancing battery safety during use. A major obstacle was tackling structural imperfections in the anode material, such as dendrite growth and needle-like formations that developed during charging and discharging cycles, causing short circuits and failures in devices [17].

As an alternative to metallic lithium, graphite has gained an important place in the batteries field, being a form of carbon with a layered structure. Graphite became the most widely used anode material due to outstanding properties, such as stable structure, good electrical conductivity and the ability to intercalate lithium ions effectively. Based on research from scientific literature, graphite has been demonstrated to be a suitable anode for LiB with high storage capacity. Graphite has established itself as the predominant anode material for lithium-ion batteries (LIBs), maintaining a 98% market share due to its cost-effectiveness, abundance, high energy and power density, and long cycle life [18]. Despite extensive research on alternative high-capacity anode materials such as silicon [19], silicon oxides, lithium metal [20], sulfides and oxides, none have yet demonstrated a sufficiently competitive performance for widespread commercial adoption. Additionally, patent and publication trends suggest that while fundamental research on graphite is limited, its industrial relevance remains significant [21].

Despite its widespread use, graphite has certain limitations. Its theoretical lithium storage capacity is relatively low, restricting the maximum energy a battery can store. Moreover, the repeated intercalation and deintercalation of lithium ions can induce structural modifications over time, contributing to gradual performance degradation [22].

Currently, the reigning champion for anode material is graphite, a form of carbon with a layered structure. Its ability to reversibly intercalate (nestle) lithium ions between its layers makes it ideal for storing and releasing them efficiently. However, graphite has limitations. Its theoretical capacity for storing lithium is relatively low, meaning the battery cannot hold as much energy as possible. Additionally, the intercalation process can cause structural changes over time, leading to performance degradation [23].

2.2.2. Silicon

Silicon is thought to be a promising anode material for increasing battery performance. In general, silicon anode materials are mainly composed of silicon nanowires, silicon nanospheres, silicon nanoparticles and silicon films [24–26].

The theoretical capacity of Si is 3590 mAhg^{-1} resulting from the analysis of the fully alloyed form of $\text{Li}_{15}\text{Si}_4$ at room temperature. This property increases to 4200 mAhg^{-1} at elevated temperatures and corresponds to the formation of $\text{Li}_{22}\text{Si}_5$ compound. The significant increase in capacity is explained by the ability of silicon to alloy with lithium and form Li_xSi compounds during the lithiation process. This high capacity together with the volumetric capacity places silicon at the top of the anode materials [27]. Although silicon-based anodes have many advantages, such as high theoretical capacity and low price, their large-scale use is hindered by two major problems: charge-induced volume expansion and lack of stability of their interface with the electrolyte, referred to as solid electrolyte interface (SEI) propagation. In addition, the significant volume fluctuations experienced by silicon-based anodes during discharge/charge processes promote the formation of undesirable solid electrolyte interfaces (SEI) [28]. In recent years, the failure mechanisms of silicon used as anode materials have been intensively studied to improve battery performance [29]. One strategy focuses on creating hybrid nanostructured anodes, such as silicon nanoparticles attached to vertically aligned carbon nanotubes (VACNTs) [30]. This design accommodates silicon's volume fluctuations, enhancing electrode stability and delivering a reversible capacity of 1330 mAh/g over 2000 cycles [31].

An additional approach involves enclosing silicon nanoparticles in carbon shells, forming a buffer that allows expansion and reduces structural damage. This method has shown enhanced cycle life, electrical conductivity and capacity stability [32].

Silicon nanowires have also been investigated as potential anode materials in recent years. These nanowires diffuse volumetric expansion more efficiently, which results in lower mechanical stress and better preservation of anode integrity. Some modeling investigations have analyzed the stress concentration within these structures and provided guidance for creating stronger silicon anodes [33].

Even with these advancements, problems like the formation of a solid-electrolyte interphase (SEI) and changes in capacity that cannot be recovered continue to exist. Current research is directed towards improving silicon-based anode designs aimed at high energy density lithium-ion batteries by performing surface treatment of electrodes and decreasing the energetics of the electrode [34].

2.2.3. Lithium Titanate ($\text{Li}_4\text{Ti}_5\text{O}_{12}$, LTO)

Another anode material used predominantly in the manufacture of Li-ion batteries is lithium titanate ($\text{Li}_4\text{Ti}_5\text{O}_{12}$, LTO). The property of this type of material to have a lithium intercalation potential of approximately 1.55 V (higher than Li^+/Li) gives it at the same time a special advantage that improves the safety of the battery and implicitly of its users. Moreover, LTO presents a “zero deformation” feature, which implies minimal volume variation during the lithiation/delithiation processes and at the same time contributes to an improvement in the life cycle and structural integrity [35]. Another strong point of LTO is the activation speed of charging/discharging processes. Besides all advantages that LTO presents, its limited Li diffusion coefficient and electronic conductivity restrict its uses. Although LTO anodes demonstrate excellent characteristics during rapid charging, this impacts the growth of Li dendrites and the development of solid electrolyte interphase (SEI) films. Furthermore, LTO's applications are constrained by its low theoretical capacity (around 175 mAh g^{-1}) and its low electrical conductivity (approximately $10^{-13} \text{ S cm}^{-1}$) [36].

To address the issues, multiple strategies have been used, such as nanostructuring to reduce lithium-ion diffusion distances, introducing higher-valence cations to boost electronic conductivity and adding conductive carbon-based materials to enhance overall electrochemical performance [37]. For example, carbon coatings on LTO improve electrical conductivity and expand the active surface for lithium-ion interaction, playing a key role in preserving capacity during rapid charge/discharge cycles. For example, carbon coatings on LTO enhance specific capacity (206 mA h g^{-1} at 20C being appropriate to LTO without carbon layers which was 224 mA h g^{-1} at 1C) and increase the active area for lithium-ion interaction, which is crucial for maintaining capacity during fast charge/discharge cycles [38].

E.J. Moon et al. introduce a new method for improving the electrochemical efficiency of lithium titanate ($\text{Li}_4\text{Ti}_5\text{O}_{12}$, LTO) anodes through the application of a nanometer-thick layer of partially lithiated titania (Li_xTiO_2 , where x is approximately 0.5). This coating of mixed ionic-electronic conductor (MIEC) greatly enhances the transport of lithium ions and electrons, resulting in improved rate capability and capacity retention, thereby making LTO anodes better fit for high-power lithium-ion battery uses [36,39].

These innovations have established LTO as a practical substitute for conventional graphite anodes, especially in uses that demand high safety and extended lifespan, like electric vehicles and large-scale energy storage systems [40].

2.2.4. Tin (Sn)

Tin (Sn) and its compounds have emerged as potential candidates for Li-ion battery applications, attracting considerable interest in recent years. The compound $\text{Li}_{4.4}\text{Sn}$, resulting from the alloying of pure tin with lithium, exhibits a high theoretical capacity ($\sim 944 \text{ mAh/g}$) compared to conventional graphite anodes ($\sim 372 \text{ mAh/g}$) [41]. In addition, the operation of tin-based anodes at a moderate working potential ($\sim 0.4\text{--}0.8 \text{ V}$) reduces the risk of lithium dendrite formation and provides advanced safety in battery use [42].

However, the main challenge for Sn anodes is the volume expansion ($\sim 300\%$) during lithiation, which leads to decreased electrical contact, electrode pulverization or rapid capacity fading [43]. To overcome this limitation, recent studies have focused on nanostructured tin materials, innovative composite designs and doping techniques [44].

Over time, research in the field has aimed to improve the properties of Sn-based anodes by developing new strategies. Duan et al. [45], demonstrate in their study that the design of Sn nanoparticles is very important, since hollow structures and 2D nanomaterials significantly influence the adaptation of materials to volume changes and cyclic stability. For example, SnO_2 nanowires show improved electrochemical performance and cycle life due to short lithium diffusion paths and large contact surface area [43].

In the same study, Wang et al. [43] suggest that coating tin with carbon-based materials (graphene, CNTs or amorphous carbon) improves electronic conductivity and reduces mechanical stress. Sn-C composites have demonstrated superior properties in terms of lithiation/delithiation rate and cycle life compared to pure Sn anodes due to the ability to mitigate large volume changes.

Doping tin-based anodes with elements such as Fe, Co or Cu has been shown to improve structural stability and conductivity. A study by Zhang et al. [42] shows that Sn-Cu and Sn-Fe alloys exhibit improved performance in terms of lithiation/delithiation cycling and structural integrity by suppressing phase segregation and mitigating volume fluctuations. These superior characteristics ultimately lead to improved battery lifetime.

Recent studies on MOF-derived Sn-based anodes highlight the high potential due to the presence of porous structures, which effectively alleviate the volume expansion stress while maintaining high-capacity retention. These types of materials improve the

electrochemical performance of batteries, offering a promising development path for next-generation Li-ion batteries [44].

Although there have been significant advancements, additional research is essential to enhance the long-term cycling stability and scalability of Sn-based anodes. Issues like considerable initial capacity loss, inconsistent solid electrolyte interphase (SEI) formation and high manufacturing costs need to be tackled before these can be commercially adopted. Future research should concentrate on multi-dimensional hybrid structures, sophisticated surface modifications, and electrolyte engineering to improve the electrochemical performance of Sn-based anodes [42,43].

2.2.5. Carbon Nanotubes (CNTs) and Graphene

Carbon nanotubes (CNTs) and graphene have attracted attention as high-potential materials for the fabrication of anodes in Li-ion batteries due to their exceptional properties, such as high conductivity (106 Sm^{-1} at 300 K for single-walled CNTs (SWCNTs) and $>105 \text{ Sm}^{-1}$ for multi-walled nanotubes (MWCNTs)) [46], mechanical tensile strength (13 to 53 GPa) [47] and large surface area (more active sites for lithium-ion storage and faster charge transfer). Some of the advantages of carbon nanostructured anodes compared to traditional graphite anodes include enhanced theoretical capacity, improved charge and discharge rate performance and better cycle life stability [48].

Studies on carbon nanotubes (CNTs) and especially MWCNTs as anode materials have demonstrated superior electrochemical performance. MWCNTs exhibit a discharge capacity of approximately 312 mAh/g. Chemical etching can further increase lithium storage capacities, with etched MWCNTs reaching values up to 2087 mAh/g [49,50].

Graphene, represented as a monolayer of carbon atoms with sp^2 bonds can offer remarkable properties as an anode material, such as electrical conductivity or mechanical flexibility [51]. This type of material provides a theoretical specific capacity of 744 mAh/g based on Li_2C_6 , which is twice that of graphite (372 mAh/g) [52], and a high specific surface area ($2630 \text{ m}^2/\text{g}$), which represents an advantage in lithium-ion diffusion and storage [53]. Integrating graphene into anode materials has been demonstrated to improve the overall electrochemical performance of LIBs [54].

The use of CNTs and graphene as anode materials in Li-ion batteries faces challenges such as potential agglomeration, restacking or complex synthesis processes. Future research aims to develop scalable fabrication methods, optimize electrode design and improve the properties of these nanomaterials for successful use in commercial applications [55].

2.2.6. High Entropy Oxides (HEOs)

A relatively new class of materials has attracted researcher attention in the battery field: high entropy oxides (HEOs). These oxides are based on high entropy alloy (HEA) principles, and they contain five or more metal cations in near-equimolar ratios. HEOs form single-phase crystal structures stabilized by high configurational entropy. The first HEO, $(\text{MgNiCuCoZn})_{0.2}\text{O}$ in a rock salt structure, was reported in 2015 by Rost et al. [56]. The study demonstrates that this entropy-driven design promotes a reversible transformation between multi-phase and single-phase states, with random and homogeneous cation distributions. These research attest that the concept of configurational disorder can be a strategic pathway for discovering new crystalline phases and engineering materials with tailored properties [56].

The advantages of their individual constituents ensure a remarkable improvement in a wide range of technologies, such as energy storage systems, thermal barriers, or catalysts [57]. In the field of lithium-ion batteries, these materials have the potential to enhance current capabilities due to their advantageous characteristics, including better cycling

stability and enhanced electrochemical performance [58]. Researchers have synthesized spinel HEO nanoparticles, $(\text{Cr}_{0.2}\text{Mn}_{0.2}\text{Co}_{0.2}\text{Ni}_{0.2}\text{Zn}_{0.2})_3\text{O}_4$, using a solution combustion method. This HEO demonstrated excellent cycling stability, maintaining a reversible capacity of $132 \text{ mAh}\cdot\text{g}^{-1}$ after 100 cycles at $100 \text{ mA}\cdot\text{g}^{-1}$ and $107 \text{ mAh}\cdot\text{g}^{-1}$ after 1000 cycles at $1 \text{ A}\cdot\text{g}^{-1}$. The high entropy effect contributes to the structural stability during electrochemical cycling, enhancing the material's performance as an anode in LIBs [59]. On the other hand, the porous spinel-phase HEO, $(\text{Cr}_{0.2}\text{Fe}_{0.2}\text{Co}_{0.2}\text{Ni}_{0.2}\text{Zn}_{0.2})_3\text{O}_4$, was synthesized at low temperatures using a sol-gel method. In this context, it has been observed that a porous structure facilitates transport and accommodates volume change during cycling. This material delivered a capacity of $1022 \text{ mAh}\cdot\text{g}^{-1}$ after 1000 cycles at $1 \text{ A}\cdot\text{g}^{-1}$ and maintained $220 \text{ mAh}\cdot\text{g}^{-1}$ even at an ultrahigh current density of $30 \text{ A}\cdot\text{g}^{-1}$. The entropy stabilization effect suppresses cation short-range order, ensuring fast lithium-ion transport and excellent electrochemical performance [60]. Nan Qiu et al. [61] examined the behavior and properties of a $(\text{Mg}_{0.2}\text{Co}_{0.2}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{Zn}_{0.2})\text{O}$ mixture. In comparison to graphite, which has a storage capacity of approximately 372 mAh g^{-1} , the studied material demonstrated a significant enhancement, achieving a value of 700 mAh g^{-1} during the conversion reaction from metal oxides to their metallic forms. During this process, certain components are expected to function as electrochemically active groups, thereby enhancing the overall capacity, while other elements remain electrochemically inactive, helping to improve the structural stability for prolonged cycling [62].

Table 1 presents a comparative overview of key performance indicators for several anode materials used in lithium-ion batteries, including graphite, silicon, lithium titanate (LTO), tin, carbon nanotubes (CNTs), graphene and high entropy oxides (HEOs). These materials are evaluated based on their specific capacity, energy density, power density, cycling stability, material cost and number of charge–discharge cycles. Quantitative values are provided where consolidated data are available in the literature; for the remaining entries, the qualitative classifications follow the explicit criteria defined in the notes below the table.

Table 1. Comparative overview of key performance metrics for selected anode materials used in lithium-ion batteries.

Anode Material	Specific Capacity, (mAh/g)	Energy Density, Wh/L ¹	Power Density ²	Stability	Cost	Life Cycle	Ref.
Graphite	~372	~600	Medium	High	Low	Medium	[63,64]
Silicon	~4200	~1000	High	Low	High	Medium	[65,66]
Lithium Titanate (LTO)	~175	>177	High	High	High	High	[67,68]
Tin and compounds	~944	High	Medium	Low	Low	-	[41,69]
Carbon Nanotube (CNTs)	~300–1000	High	Excellent	High	High	High	[52,70]
High entropy oxides	~200–1000	High	High	High	High	High	[61,71]

¹ For the emerging materials (tin-based compounds, CNTs and HEOs), standardized practical volumetric energy densities are not yet consolidated in the literature; the qualitative labels are therefore assigned relative to the electrode-level energy density of commercial graphite (~600 Wh/L), on the basis of the theoretical and practical specific capacities reported in the cited references: “High”—expected to exceed the graphite benchmark (>600 Wh/L); “Medium”—comparable to graphite (300–600 Wh/L); “Low”—below 300 Wh/L. ² Power density is classified according to the rate capability reported in the cited references: “Excellent”—stable capacity retention at rates of 10C or higher; “High”—stable operation at 2–10C; “Medium”—stable operation only at rates below 2C.

Graphite, the conventional anode material, offers excellent stability and low cost but suffers from limited specific capacity (~372 mAh/g). In contrast, silicon exhibits an exceptionally high theoretical capacity (~4200 mAh/g), but it faces critical challenges related to stability and life cycle. Lithium titanate provides superior power density and stability but has lower energy density and specific capacity. Tin-based materials offer energy density higher than graphite and enhanced specific capacity, though issues like

power density and stability remain obstacles. To enhance the performance of carbon nanostructure-based anodes (such as CNTs and graphene), future research should focus on improving their specific capacity while maintaining their excellent electrical conductivity and structural stability. High entropy oxides (HEO), a relatively new class of materials, demonstrate promising multi-element synergy effects that may enable stable long-term cycling and moderate capacities, though they are still in the early stages of development and optimization [72].

This comparative assessment helps identify trade-offs and guide the selection or engineering of anodes tailored for specific performance requirements in applications ranging from portable electronics to electric vehicles and grid storage.

3. Synthesis Methods for Anode Materials Used in Li-Ion Batteries

The selection of the anode material and particularly the synthesis method employed in its preparation is a critical factor in the development of advanced Li-ion batteries with enhanced performance [3]. Synthesis methods not only affect the structural and morphological properties of the anode but also influence the distribution of active particles, degree of crystallinity, specific surface area, and overall electrochemical behavior. For this reason, an in-depth analysis of existing synthesis technologies is essential for optimizing battery performance and adapting materials to the diverse requirements of modern applications.

Based on the physical and chemical processes involved, the synthesis methods for anode materials can be broadly classified into several categories presented in Figure 2.

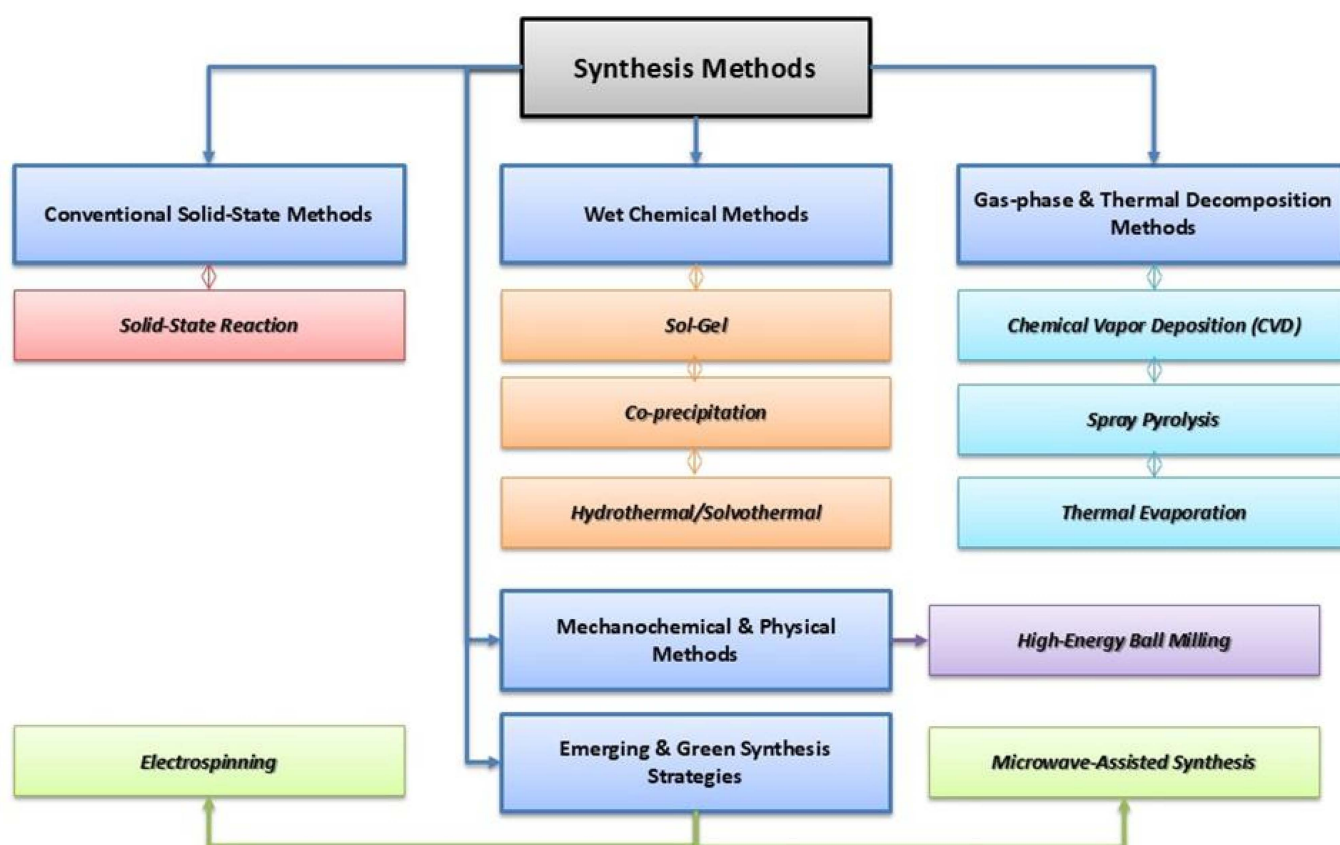


Figure 2. Classification of the synthesis methods for anode materials for Li-ion batteries.

The following section will provide a comprehensive literature review of the main synthesis methods employed for producing anode materials, outlining their advantages,

limitations, and the influence of each technique on the structural, morphological and electrochemical characteristics of Li-ion batteries.

3.1. Solid-State Reaction

The solid-state reaction method is one of the most traditional and widely used synthesis techniques for producing anode materials for Li-ion batteries. It typically involves the high-temperature calcination of solid precursors (often metal oxides, carbonates, or other inorganic salts) in a stoichiometric ratio, followed by grinding and repeated heat treatments to ensure homogeneity and complete phase formation. This technique is especially suitable for synthesizing metal oxides and composite materials due to its simplicity, cost-effectiveness, and scalability for industrial applications. However, its key limitation lies in the poor control over particle size and morphology, which may lead to large grain sizes, low specific surface area, and limited lithium-ion diffusion pathways [73]. Despite these limitations, the solid-state route remains relevant, especially for fabricating stable and structurally simple materials such as $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO) [74], SiO-based anodes [75] or Fe_3O_4 -based anodes. For instance, LTO synthesized via solid-state reaction often exhibits excellent cycling stability due to its zero-strain property and high structural integrity [74]. Additionally, composite materials such as $\text{SnO}_2\text{-C}$ or $\text{Fe}_3\text{O}_4\text{-C}$ have also been prepared by this method, where subsequent carbon coating improves their conductivity and cycling behavior.

To enhance the properties of materials synthesized via solid-state methods, researchers have explored process optimizations such as precursor modification, the use of fluxes, controlled atmosphere treatments, and post-synthesis grinding to reduce agglomeration and improve homogeneity [76,77].

Overall, while solid-state reactions are limited in terms of morphological control, their reliability and scalability make them a valuable approach for the synthesis of anode materials, especially when combined with secondary treatments or hybrid techniques.

3.2. Sol–Gel Synthesis

The sol–gel method is a versatile and widely used wet-chemical synthesis technique for fabricating advanced materials, including anode materials for lithium-ion batteries. It involves the hydrolysis and condensation of metal alkoxides or inorganic salts to form a colloidal solution (sol), which gradually evolves into a gel-like network through polymerization and drying processes. Upon thermal treatment, the gel can be converted into a crystalline or amorphous oxide material, depending on the application. One of the main advantages of the sol–gel process is the precise control over composition and microstructure, allowing the synthesis of homogeneous nanostructured materials with high surface area and porosity that are particularly beneficial for improving lithium-ion diffusion and electrochemical performance of anode materials. This method also enables the preparation of doped or composite materials by simply adjusting the precursor chemistry, which is useful for tailoring the electronic conductivity and capacity of the anodes. Moreover, the relatively low synthesis temperatures of the sol–gel method help preserve nanostructures and prevent agglomeration, which is essential for achieving good electrochemical performance.

In their work, Yin et al. [78] present the sol–gel synthesis of Al-doped, carbon-coated $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO) anodes, leveraging the method's ability to produce homogenous, nano-sized particles with uniform doping and carbon coating. The resulting material exhibited enhanced Li^+ diffusion, discharge capacity up to 174.1 mAh/g at 1C (the C-rate is calculated by dividing the charge or discharge current by the theoretical capacity of the battery), and excellent cycling stability (97% after 100 cycles at 20C), confirming the sol–gel method as an effective route for advanced anode materials. Another study, published in 2021 by Thauer and his co-workers [79], reports the sol–gel synthesis of $\text{Li}_3\text{VO}_4/\text{carbon}$ ($\text{Li}_3\text{VO}_4/\text{C}$)

composites for use as anode materials in lithium-ion batteries. Their findings led to a significant improvement in electrochemical performance: the optimized composite delivered a reversible capacity of 366.3 mAh/g at 0.2C, maintained 270.6 mAh/g after 200 cycles at 1C, and showed excellent rate capability (175.3 mAh/g at 5C).

However, the sol–gel process can be limited by factors such as long processing times, the use of expensive precursors (metal alkoxides), and the need for careful control of processing parameters like pH, temperature, and solvent composition. Despite these challenges, sol–gel remains a promising and adaptable route for producing advanced anode materials with enhanced lithium storage capabilities.

3.3. Hydrothermal/Solvothermal Synthesis

Hydrothermal and solvothermal synthesis methods are widely employed for the preparation of nanostructured anode materials due to their ability to produce crystalline powders with controlled morphology, uniform particle size, and high phase purity under relatively mild conditions. These methods involve the chemical reaction of precursors in a sealed autoclave at elevated temperatures (typically 100–250 °C) and autogenous pressures, using water or organic solvents as the reaction medium. The hydrothermal process is particularly suitable for synthesizing complex oxides and hierarchical structures, such as nanorods, nanoflowers, or hollow spheres, which can offer improved lithium-ion transport pathways and structural stability during cycling. The low synthesis temperature also limits grain growth, allowing for the retention of high surface areas and active sites.

Hydrothermal synthesis has been successfully used to prepare anode materials like TiO_2 , Fe_3O_4 , MnO_2 , and $\text{Li}_4\text{Ti}_5\text{O}_{12}$, often exhibiting enhanced rate capability and cycling performance due to the resulting tailored nanostructures. Courtel et al. [80] reported the hydrothermal synthesis of ultra-fine ZnMn_2O_4 nanoparticles (≤ 5 nm) as high-performance anode materials for Li-ion batteries. The hydrothermal method, using ethylene glycol as a solvent enabled precise control over particle size and prevented agglomeration, yielding highly uniform nanoparticles without the need for post-synthesis annealing. Electrochemical testing showed a stable reversible capacity of 430 mAh/g after 100 cycles at C/10. The improved cycling stability and performance were attributed to the nanoscale dimensions, which facilitated lithium diffusion and enhanced the reversibility of conversion reactions, as well as the use of a lithium carboxymethylcellulose (LiCMC) binder to mitigate the effects of volume expansion. A few years later, in 2019, Li et al. [81] reported in their study the synthesis of nitrogen-doped TiO_2 /reduced graphene oxide (N- TiO_2 /rGO) hybrids via a hydrothermal approach, employing thiourea as the nitrogen source. The hydrothermal method enabled uniform doping and intimate integration of TiO_2 with rGO, promoting the formation of N–Ti–O and N–C bonds. These structural features resulted in a higher specific surface area and a greater number of active lithium storage sites. Consequently, the N- TiO_2 /rGO hybrid demonstrated excellent electrochemical performance and cycling stability, retaining 210 mAh/g after 2000 cycles. The use of hydrothermal synthesis proved particularly effective for achieving controlled doping and enhancing the overall electrochemical behavior of the anode material.

Solvothermal methods, which use organic solvents instead of water, offer even finer control over particle morphology and can facilitate the formation of metastable phases or carbonaceous coatings when using carbon-containing solvents. These methods are especially useful for the synthesis of composite or hybrid anode materials. For example, Shen et al. [82] obtained ZnO-loaded porous carbon (ZnO/PC) composites for the first time via a facile solvothermal method. This in situ approach ensured a uniform dispersion of ZnO nanoparticles within the porous carbon matrix, facilitating strong interfacial contact and improved structural stability. The porous carbon host effectively accommodated the

volume changes in ZnO during charge–discharge cycles, mitigating electrode degradation and contributing to superior cycling performance (high reversible capacity of 653.7 mAh/g after 100 cycles).

Despite their advantages, hydrothermal and solvothermal techniques may be limited by long reaction times, low yield, and the need for high-pressure equipment, which can affect scalability. Nevertheless, their versatility and effectiveness in producing well-defined, high-performance nanomaterials make them attractive for next-generation lithium-ion battery anodes.

3.4. Co-Precipitation Synthesis

Co-precipitation is a widely used wet-chemical synthesis method for preparing anode materials due to its simplicity, cost-effectiveness, and ability to produce chemically homogeneous multicomponent powders. In this process, metal cations are simultaneously precipitated from an aqueous solution (usually by the addition of a base such as NaOH or NH_4OH) forming a uniform precursor that can be thermally treated to yield the desired oxide or composite phase.

One of the main advantages of co-precipitation is its ability to achieve molecular-level mixing of multiple metal ions, which is particularly beneficial for the synthesis of complex oxides or doped materials. The method also offers good control over stoichiometry, relatively low processing temperatures, and compatibility with scalable production, making it attractive for industrial applications. It has been successfully applied to synthesize a variety of anode materials, such as Fe_3O_4 , Ni-based oxides [83], as well as composite materials like $\text{Fe}_3\text{O}_4/\text{C}$, $\text{Li}_4\text{Ti}_5\text{O}_{12}$ with doping elements [84,85] or CuO-NiO/rGO [86]. For example, Gu et al. [85] used in their work the co-precipitation synthesis to obtain MgCo_2O_4 -coated $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO) composite anodes for lithium-ion batteries. The method enabled uniform deposition of the MgCo_2O_4 spinel layer on the LTO surface, enhancing electronic conductivity and structural stability. Owing to the precise stoichiometric control and homogeneous coating achievable through co-precipitation, the composite exhibited improved electrochemical performance, including higher specific capacity and rate capability (300 mAh/g, two times higher than that achieved by pristine LTO), confirming the method's effectiveness for surface-modified anode design.

However, some limitations exist. Co-precipitation is sensitive to pH, temperature, and mixing rate, which can lead to particle agglomeration or compositional inhomogeneity if not well controlled.

3.5. Spray Pyrolysis Synthesis

Spray pyrolysis is an efficient and scalable technique for synthesizing fine, spherical particles of anode materials for lithium-ion batteries. The method involves atomizing a precursor solution into fine droplets, which are then transported by a carrier gas through a heated reactor. Upon heating, the solvent evaporates and the precursor undergoes thermal decomposition, resulting in the formation of solid particles with controlled morphology and composition. One of the key advantages of spray pyrolysis is its ability to produce uniformly sized, dense, and spherical particles in a single-step continuous process. Additionally, multi-component systems or doped materials can be easily synthesized by dissolving all precursors in the initial solution, leading to excellent chemical homogeneity in the final product.

Spray pyrolysis has been successfully employed to prepare various anode materials such as $\text{Li}_4\text{Ti}_5\text{O}_{12}$ [87,88], SnO_2 , Fe_2O_3 [89], and TiO_2 -based composites. For instance, Birrozzi et al. [90] explored the use of flame spray pyrolysis (FSP) for the synthesis of $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO) nanoparticles, aimed at enhancing high-power performance in lithium-

ion battery anodes. Their study reports that FSP can produce LTO nanoparticles with an average particle size of around 30–50 nm, offering high surface areas ($\sim 100 \text{ m}^2/\text{g}$) and excellent electrochemical properties, such as a discharge capacity of approximately 160 mAh/g.

However, some challenges include high energy consumption, potential precursor loss, and difficulty in controlling complex morphologies without post-treatment. Additionally, the reactor setup requires precise optimization of parameters such as droplet size, temperature profile, and residence time to ensure consistent material quality.

3.6. Chemical Vapor Deposition

Chemical vapor deposition (CVD) is a widely used technique for synthesizing high-quality anode materials in lithium-ion batteries. In CVD, gaseous precursors are introduced into a chamber, where they react on a substrate at elevated temperatures to form thin, uniform films. This process allows precise control over the material's composition, thickness, and morphology, which is crucial for optimizing anode performance.

A key advantage of CVD is its ability to create conformal coatings that help stabilize the anode during the significant volume changes typical of silicon-based materials. These coatings enhance cycling performance and prevent structural degradation. Additionally, CVD enables fine-tuning of properties like porosity and crystallinity, optimizing ion diffusion and conductivity for better battery performance. For example, Laurenti et al. [91] demonstrate in their work the chemical vapor deposition growth of ZnO nanostructures directly onto stainless steel current collectors for use as binder-free anodes in lithium-ion batteries. The CVD method enabled the formation of well-aligned, polycrystalline nanostructures with a high surface area and strong adhesion, resulting in excellent electrode/electrolyte contact and efficient charge transport. Despite the absence of additives, the electrode delivered a stable specific capacity of $\sim 400 \text{ mAh/g}$ with high Coulombic efficiency and over 62% capacity retention after prolonged cycling. In another work, Zhu et al. [92] used CVD to deposit a uniform carbon coating on rutile TiO_2 , effectively enhancing its conductivity and structural stability. The controlled carbon layer, optimized via secondary annealing, mitigated volumetric changes during cycling and improved electrochemical performance. The resulting $\text{TiO}_2\text{-C}$ composite exhibited high capacity retention over 200 cycles and excellent rate capability, highlighting CVD's potential for tailoring surface properties and boosting anode performance in Li-ion batteries [93,94].

However, challenges such as the high temperatures required for the process and the need for specialized equipment can limit scalability and increase production costs.

3.7. Emerging and Green Synthesis Method

Recent advancements in materials science have led to the development of emerging and environmentally friendly synthesis methods aimed at improving the performance, sustainability, and scalability of lithium-ion battery anodes. These techniques offer energy-efficient processing, reduced environmental impact, and often allow for precise nanostructuring or the use of biomass-derived resources.

3.7.1. Electrospinning

Electrospinning is a promising technique for the fabrication of one-dimensional nanofiber materials used as anode structures in lithium-ion batteries. It involves the application of a high-voltage electric field to a polymer-based precursor solution, which leads to the formation of ultrafine fibers.

This method offers several advantages for battery electrode design. First, the resulting nanofibrous morphology provides a large surface area, short diffusion paths for Li^+ ions, and continuous electron transport pathways, which are highly beneficial for improving rate

performance and specific capacity. Second, electrospun fibers can be easily modified by incorporating active metal oxides (SnO_2 , TiO_2 , Fe_3O_4), carbon sources, or dopants, either in the spinning solution or via post-treatment processes such as thermal carbonization or reduction. Also, it allows for the production of binder-free, flexible electrodes, which are advantageous for next-generation lithium-ion battery technologies, including flexible or wearable devices. Moreover, by adjusting parameters such as voltage, solution composition, flow rate, and collector distance, the fiber diameter, porosity, and surface characteristics can be tailored to optimize electrochemical behavior.

Numerous studies have reported high-performance electrospun anodes. For instance, Kim et al. [95] obtained Ni-doped SnO_2 –carbon nanofiber (CNF) composites via electrospinning and carbonization, resulting in uniformly distributed 15 nm SnO_2 /NiO nanoparticles embedded on conductive CNFs. The electrospinning technique enabled the formation of binder-free, porous, and highly interconnected nanostructures with enhanced Li^+ diffusion and volume-buffering effects, which delivered a high reversible capacity of 447.6 mAh/g after 100 cycles. In another example, Fe_3O_4 /C composite fibers were also synthesized by the electrospinning method, followed by carbonization in a vacuum and reported by Gu and his co-workers [96]. The resulting nanostructures exhibited a high discharge capacity of 508.2 mAh/g after 100 cycles and a strong rate capability of 160 mAh/g at 5 A/g.

Despite its many advantages, electrospinning has some challenges, including relatively low throughput compared to other methods, the need for post-sintering or stabilization steps, and potential issues with fiber uniformity at scale.

3.7.2. Elaboration in the Induction Furnace Followed by Oxidation in the Attritor—A Hybrid Approach for Synthesizing Complex Oxides

A relatively new and promising route for synthesizing advanced anode materials involves a hybrid process that combines induction furnace melting with controlled oxidation in an attritor mill. This method allows for the efficient preparation of nanostructured complex oxides with well-controlled morphology and composition, suitable for lithium-ion battery applications.

The process begins with the melting of high-entropy alloys in an induction furnace, where heat is generated via electromagnetic induction. An induction furnace typically consists of a power supply, an inductor coil, a capacitor, and a crucible. When current flows through the inductor, it generates a magnetic field, where the magnetic lines of force concentrate in the center of the coil and loop outward around it (Figure 3). This setup enables efficient induction heating of conductive materials placed within the coil. Thanks to the electromagnetic stirring effect, both the temperature and chemical composition of the molten alloy are more uniformly distributed, ensuring better homogeneity. Compared to arc melting, induction melting offers higher energy efficiency and avoids contamination issues caused by electrode materials. Moreover, it eliminates the risk of gas dissociation at high temperatures and the subsequent absorption of gases into the melt—problems commonly associated with other high-temperature melting methods [97].

However, this same stirring effect can increase the interaction between the molten material and the furnace lining, potentially leading to gas absorption and unstable alloy properties. Therefore, careful process control is essential to minimize these effects and ensure high-quality material output.

This technique enables rapid and uniform heating, high thermal efficiency, and excellent compositional homogeneity, while reducing contamination risks due to the absence of direct contact with a heating source. Operating in an inert atmosphere (argon or nitrogen) and at temperatures from 1200 to 1500 °C, the process ensures full melting and homogeneous alloy formation [98].

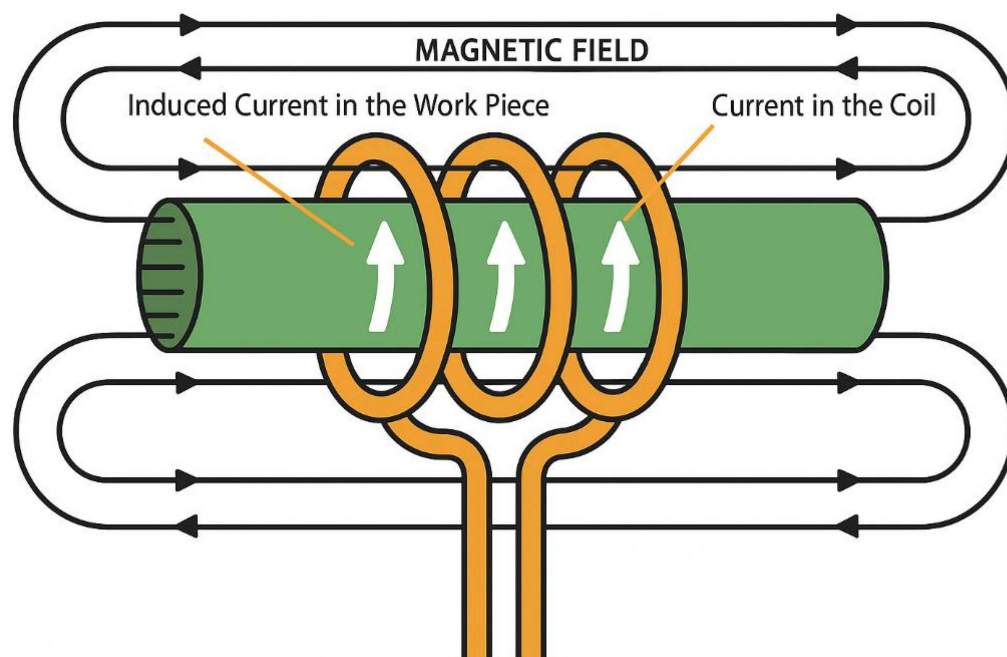


Figure 3. Diagram of magnetic fields and eddy currents in the induction furnace.

In the second step, the pre-formed ingots are subjected to high-energy mechanical milling in an attritor mill, carried out in a controlled oxidative atmosphere (air or oxygen). This promotes gradual and uniform oxidation of the metallic alloy, facilitated by repeated impacts between grinding balls and the material. The result is the formation of nanostructured complex oxides with fine particle sizes (often in the 1–100 nm range) and enhanced surface area, which are critical for improving electrochemical activity and Li^+ storage behavior. The final material's performance is highly influenced by operational parameters such as milling time (5–50 h), rotation speed (200–500 rpm), atmosphere and the ball-to-powder ratio. The process can be tuned to optimize particle size, degree of oxidation and structural integrity.

Importantly, this approach supports environmental sustainability, as it is compatible with the use of recycled raw materials, aligning with green chemistry principles and circular economy goals. The resulting materials exhibit a balance of high thermal and chemical stability and enhanced Li-ion storage capacity, making them excellent candidates for next-generation anode materials.

Therefore, the development of advanced anode materials for Li-ion batteries has significantly benefited from a wide variety of synthesis methods, each offering unique advantages in terms of material properties, performance and scalability. Methods such as sol-gel, hydrothermal and chemical vapor deposition have allowed for precise control over the morphology, porosity and crystallinity of anode materials, which directly influence their electrochemical performance, including capacity, stability and cycling efficiency (Table 2).

While significant progress has been made, there is still a need for further exploration into cost-effective, scalable, and environmentally sustainable synthesis routes. Ultimately, the continuous development and optimization of synthesis techniques will be key to advancing the performance and widespread adoption of Li-ion battery technology, driving forward progress in energy storage systems for a variety of applications.

Table 2. Unified comparison of the synthesis methods discussed in this review in terms of scalability, relative cost, energy consumption, representative electrochemical performance and environmental potential (the detailed advantages and limitations of each method are presented in Section 3).

Synthesis Method	Scalability	Relative Cost	Energy Consumption	Representative Electrochemical Performance	Environmental Potential
Solid-state reaction	High	Low	High (prolonged high-temperature calcination)	Li ₄ Ti ₅ O ₁₂ with excellent cycling stability, owing to its zero-strain behavior [74]	Medium
Sol-gel	Medium	Medium to high (alkoxide precursors)	Medium (low synthesis temperature, but drying and calcination required)	Al-doped, carbon-coated Li ₄ Ti ₅ O ₁₂ : 174.1 mAh/g at 1C and 97% capacity retention after 100 cycles at 20C [78]	Medium
Hydrothermal/solvothermal	Medium (batch autoclaves)	Medium	Low to medium (100–250 °C, long reaction times)	N-doped TiO ₂ /rGO: 210 mAh/g retained after 2000 cycles [81]; nMn ₂ O ₄ : 430 mAh/g after 100 cycles [80]	Medium
Co-precipitation	High	Low	Low to medium (low reaction temperature; calcination required)	MgCo ₂ O ₄ -coated Li ₄ Ti ₅ O ₁₂ : ~300 mAh/g, about twice the capacity of pristine LTO [85]	Medium
Spray pyrolysis	High (continuous, single-step)	Medium	Medium to high (continuously heated reactor)	Flame-spray-pyrolyzed Li ₄ Ti ₅ O ₁₂ nanoparticles: ~160 mAh/g with high-power capability [90]	Medium
Chemical vapor deposition	Low to medium	High (specialized equipment and precursors)	High (elevated temperatures, controlled atmosphere)	Binder-free ZnO nanostructures: ~400 mAh/g with over 62% retention after prolonged cycling [91]	Low
Ball milling	High	Low	Medium (long milling times)	In situ mixed composite and alloy anodes with improved homogeneity; the electrochemical performance depends on the subsequent treatments [76,77]	Medium
Electrospinning	Medium to low	Medium	Medium (spinning followed by carbonization/post-treatment)	Ni-doped SnO ₂ -carbon nanofibers: 447.6 mAh/g after 100 cycles [95]; Fe ₃ O ₄ /C fibers: 508.2 mAh/g after 100 cycles [96]	Medium
Induction melting + oxidation in attritor	Medium	Medium (over 80% recycled raw materials)	Medium to high (1200–1500 °C melting, but energy-efficient induction heating)	Nanostructured complex (high-entropy) oxides with enhanced Li ⁺ storage capacity and thermal/chemical stability [97,98]	Medium to high

4. Challenges in Obtaining Anode Materials with Improved Performances

Lithium-ion batteries have emerged as the predominant energy storage system across a broad spectrum of applications, ranging from portable consumer electronics to the expanding electric vehicle sector. This widespread adoption is primarily attributed to their high energy density, extended cycle life, and favorable gravimetric characteristics. Within the LIB architecture, the anode material is a critical determinant of the cell's electrochemical performance, influencing key parameters such as specific capacity, rate capability, operational lifespan, and safety [99].

As a result, there is a growing global initiative aimed at identifying and engineering alternative anode materials that exhibit higher lithium storage capacities and enhanced electrochemical performance. In parallel with efforts to optimize battery performance, increasing emphasis is being placed on the environmental implications of material synthesis and battery manufacturing processes. Conventional fabrication techniques frequently involve high energy consumption, reliance on toxic reagents, and substantial waste generation, thereby raising critical concerns regarding the long-term environmental sustainability of lithium-ion battery technologies [100]. Moreover, the limited availability of certain key

raw materials further highlights the necessity for the development of sustainable material alternatives and the implementation of efficient recycling and resource recovery strategies.

This section is organized as follows: Sections 4.1 and 4.2 address the general challenges associated with two of the most widely applied synthesis routes for anode materials, while Sections 4.3–4.6 focus specifically on the synthesis of high-entropy oxides (HEOs). This emphasis is intentional, since HEOs are the most recent class of anode materials discussed in this review and the one for which synthesis-related challenges are the least consolidated in the literature, with the simultaneous stabilization of five or more cations in a single phase amplifying the difficulties already encountered for conventional oxides. Sections 4.7 and 4.8 complete the picture by discussing the interfacial compatibility of the reviewed anode materials with solid-state electrolytes and the industrial bottlenecks that currently limit the commercialization of HEO anodes.

4.1. Combustion Synthesis

This approach frequently enables product formation and crystallization at reduced furnace temperatures relative to conventional solid-state synthesis methods. The self-propagating exothermic nature of the combustion reaction contributes to significantly shorter synthesis durations and may lead to reduced overall energy consumption [101].

Nonetheless, a major drawback of combustion synthesis lies in the pronounced sensitivity of the final material properties to a wide range of processing parameters. Key variables include the type and concentration of the fuel (e.g., urea, glycine, etc.), the selection and stoichiometry of the oxidizing agent (ranging from metal nitrates to ammonium nitrate), the heating rate employed to initiate combustion, and the composition of the surrounding atmosphere during both thermal decomposition and combustion. Even minimal deviations in these parameters can induce substantial changes in the resulting material's phase composition, crystallinity, particle size distribution, and morphological features [102]. This intrinsic sensitivity poses significant challenges to achieving consistent and reproducible synthesis of high-quality anode materials with optimized electrochemical performance.

Attaining the appropriate stoichiometric balance between the fuel and oxidizer in the precursor solution is critical for the successful execution of combustion synthesis. This precise ratio ensures the liberation of sufficient thermal energy during the exothermic reaction to enable complete conversion of precursors into the targeted oxide phase and to promote its crystallization. Inadequate quantities of either fuel or oxidizer may result in suboptimal reaction temperatures, thereby hindering the formation of the desired crystalline phase and allowing for the persistence of unreacted precursors or the generation of undesired secondary phases. Conversely, an excess of fuel may lead to incomplete combustion, producing residual carbonaceous byproducts within the final material matrix. Such impurities can adversely affect the structural integrity and electrochemical performance of the synthesized anode material [103].

In practical combustion systems, the underlying chemical reactions often exhibit greater complexity than suggested by simplified theoretical models. Secondary processes, such as the preliminary thermal decomposition of the fuel or oxidizer prior to the onset of the primary combustion reaction, can significantly influence the overall reaction dynamics. As a result, stoichiometric fuel-to-oxidizer ratios derived from idealized, balanced chemical equations may not accurately reflect the optimal conditions required for efficient combustion and desired phase formation. Consequently, empirical optimization through systematic experimentation is frequently necessary to fine-tune these ratios, ensuring favorable combustion characteristics and the successful synthesis of high-purity products.

Moreover, the gas flow dynamics within the reaction system play a pivotal role in determining the outcome of the combustion process, particularly when the precursor formulation contains a limited amount of oxidizer. Under such conditions, the reaction may become dependent on the influx of additional oxidizing species—such as atmospheric oxygen—to achieve complete fuel combustion and facilitate the formation of the desired oxide phase. Consequently, precise regulation of both the atmospheric environment and the gas flow rate is essential for maintaining optimal reaction conditions and ensuring the successful synthesis of high-purity target materials [104].

As a highly exothermic process involving elevated temperatures and potential flammable organic fuels, combustion synthesis inherently presents notable safety concerns. The rapid and potentially uncontrolled release of thermal energy can create hazardous conditions, including the risk of fire or explosion, particularly if the process is not meticulously designed and managed. Additionally, the thermal decomposition of organic fuel components during combustion can result in the evolution of various gaseous by-products, some of which pose environmental or health hazards. These emissions may include carbon dioxide (CO_2), nitrogen oxides (NO_x), ammonia (NH_3), and other nitrogen-containing species, contingent upon the specific fuel utilized [105]. For example, the thermal degradation of urea, a widely used fuel in combustion synthesis, can lead to the release of nitrogenous gases into the surrounding atmosphere, thereby necessitating appropriate containment and emission control measures.

4.2. Co-Precipitation

A major challenge inherent to the co-precipitation method is the attainment of high compositional homogeneity within the resulting precursor, particularly in the synthesis of anode materials comprising multiple metal cations. Achieving uniform and simultaneous precipitation of the constituent metal ions in the desired stoichiometric ratios across the reaction medium necessitates stringent control over a complex set of reaction parameters. These include the solution pH, reaction temperature, stirring or mixing rate, concentrations of the initial metal salt precursors, the concentration and addition rate of the precipitating agent, and the total reaction duration. Inadequate regulation of any of these parameters can lead to selective precipitation, where certain metal ions precipitate preferentially, resulting in compositional inhomogeneities within the precursor particles [106]. Such non-uniformity can be detrimental to the electrochemical consistency of the final anode material, adversely impacting its specific capacity, cycle stability, and rate performance.

Regulating the particle size, shape, and overall morphology of the co-precipitated precursor—and by extension, the final anode material post-calcination—represents another critical challenge inherent to the co-precipitation synthesis technique. The morphology of the anode material exerts a profound influence on its electrochemical behavior by affecting parameters such as the specific surface area accessible for lithium-ion intercalation, the length and tortuosity of lithium-ion diffusion pathways within the particles, and the overall packing density of the electrode. Achieving optimal morphologies—such as uniformly spherical particles with controlled porosity and a narrow particle size distribution—typically necessitates meticulous control over synthesis conditions [107]. This may include the use of tailored additives, surfactants, or structure-directing agents to modulate nucleation and particle growth dynamics. Additionally, the intrinsic tendency of precipitated particles to agglomerate or form aggregates during the reaction process poses a further complication, as such phenomena can reduce the effective surface area and adversely impact the electrochemical performance of the final anode material.

Conventional co-precipitation techniques frequently utilize chelating agents such as ammonia to regulate the solubility of metal ions in solution, thereby exerting control over

the particle size and morphology of the synthesized precursors. Nevertheless, the incorporation of ammonia in these methodologies poses substantial environmental challenges. Ammonia is a hazardous substance, and inadequate containment can result in its volatilization and subsequent atmospheric release. Furthermore, these processes typically produce considerable quantities of ammonium-laden wastewater during the filtration and washing stages, necessitating energy-intensive and economically burdensome treatment protocols prior to environmentally compliant disposal [108].

To mitigate the oxidation of metal cations—particularly transition metal ions commonly employed in lithium-ion battery (LIB) anode materials—during the co-precipitation process, it is imperative to perform the reaction under a controlled inert atmosphere, such as nitrogen or argon. The presence of oxygen within the reaction environment can induce undesirable oxidation states or promote the formation of non-target oxide phases, both of which can detrimentally affect the electrochemical performance of the resulting anode material. Ensuring an inert atmosphere throughout the co-precipitation procedure consequently increases the overall complexity and operational costs of the synthesis system.

When the co-precipitation method is applied to the recycling of spent LIBs for the recovery of valuable metals used in the synthesis of new anode materials, the process is particularly susceptible to the presence of impurities in the leachate derived from end-of-life cells. These impurities—comprising extraneous metal ions and organic residues originating from electrolyte degradation—can co-precipitate alongside the target metal species, thereby compromising the chemical purity of the recovered precursors. The incorporation of such contaminants into the precursor material may adversely affect the structural integrity and electrochemical performance of the re-synthesized anode materials [109]. Consequently, rigorous pre-treatment and purification of the leach solution are essential to ensure the removal of these impurities prior to initiating the co-precipitation step in battery recycling workflows.

4.3. Sol–Gel Synthesis of High-Entropy Oxides

While the sol–gel method provides a flexible and effective approach for synthesizing complex oxide materials with high compositional uniformity, its implementation in the fabrication of complex oxides for advanced lithium-ion battery anodes poses distinct challenges.

The hallmark of high-entropy oxides is the stabilization of a single-phase solid solution structure, primarily driven by high configurational entropy. Although the sol–gel method enables atomic-level mixing of multiple metal precursors in the solution phase, translating this homogeneity into a thermodynamically stable single-phase HEO during gelation and subsequent thermal treatment is inherently complex. The enthalpy of mixing between certain cationic species may oppose entropic stabilization, promoting the formation of secondary phases or the persistence of unreacted precursors. Consequently, high-temperature calcination is often required to kinetically facilitate single-phase formation. However, such conditions can induce undesirable grain growth and particle agglomeration, thereby compromising the electrochemical performance of the resulting material. Accurately predicting and controlling the phase evolution of these compositionally complex oxides during sol–gel synthesis remains a fundamental challenge in the development of high-performance HEO anodes [110].

Maintaining precise stoichiometry and ensuring uniform spatial distribution of five or more metal cations throughout the sol–gel process are essential for realizing the high-entropy effect and achieving optimal electrochemical performance. However, disparities in the dissolution kinetics, hydrolysis rates, and condensation behaviors of the constituent metal precursors can result in compositional gradients within the gel matrix. These inhomogeneities may be further amplified during the drying and calcination stages due to

differential shrinkage and species-specific diffusion dynamics. Achieving true atomic-level homogeneity while preventing phase segregation or preferential crystallization of individual elements remains a technically demanding task [111]. Such compositional deviations can significantly influence the resulting crystal structure, thereby impacting the structural integrity and electrochemical behavior of the material as a lithium-ion battery anode.

The sol–gel synthesis of high-entropy oxides is governed by a wide array of interdependent parameters, including the selection of metal precursors, solvent system, pH, gelling agents, drying protocols, and most critically, calcination temperature and duration. Each of these factors exerts a substantial influence on the chemical environment of the cations, the structural evolution of the gel, the crystallization dynamics, and ultimately the phase composition, particle size, morphology, and surface characteristics of the final product. The optimization of these parameters to reproducibly obtain single-phase HEOs with desirable attributes, such as nanoscale architecture, high specific surface area, or minimal structural defects, constitutes a complex, multi-variable challenge that typically necessitates extensive empirical tuning. For example, although elevated calcination temperatures may promote the formation of a stable single-phase structure, they can simultaneously induce grain coarsening and diminish surface area, both of which are detrimental to the rate capability and long-term cycling stability of lithium-ion battery anodes [110].

Another fundamental challenge in the development of high-entropy oxides for lithium-ion battery anodes is the establishment of robust correlations between the nuanced parameters of the sol–gel synthesis process and the resulting electrochemical performance. Subtle variations in phase purity, cation distribution (order/disorder), crystallite size, morphology, and defect density—all of which are highly sensitive to synthesis conditions—can profoundly influence key performance metrics such as lithium-ion diffusion kinetics, electronic conductivity, solid-electrolyte interphase formation and structural stability under repeated cycling. Elucidating these complex structure–property relationships is essential for the rational design and systematic optimization of sol–gel synthesis protocols aimed at achieving high-performance HEO-based anodes [112].

Although the sol–gel method is often regarded as relatively scalable compared to other synthesis techniques, the large-scale production of high-entropy oxides with stringent compositional control and phase purity introduces both engineering and economic challenges. The high cost of analytical-grade metal precursors, extensive solvent usage, and substantial energy requirements for drying and high-temperature calcination contribute to the overall production cost [113]. Moreover, translating the precise control over reaction kinetics, homogeneity, and phase formation observed at the laboratory scale to industrial-scale operations necessitates significant process optimization. This may include the development of continuous or semi-continuous synthesis systems to enhance scalability while preserving material quality and reproducibility.

4.4. Hydrothermal/Solvothermal Synthesis of High-Entropy Oxides

Although hydrothermal and solvothermal techniques provide promising avenues for the synthesis of nanostructured materials under relatively moderate temperature conditions, their implementation in the fabrication of HEOs for lithium-ion battery anodes remains hindered by substantial technical challenges.

The principal challenge lies in achieving the desired single-phase solid solution structure that defines HEOs. The inherently low synthesis temperatures associated with hydrothermal and solvothermal methods may not provide sufficient thermal energy to surmount the formation enthalpies of thermodynamically favored binary or ternary oxide phases, despite the stabilizing effect of high configurational entropy. Additionally, disparities in hydrolysis rates, precipitation kinetics, and solubilities among the various metal

precursors under these conditions hinder the concurrent and uniform precipitation of all constituents [114]. This often results in deviations from the targeted stoichiometry, compositional inhomogeneities within individual particles, and the emergence of undesirable secondary phases.

While these approaches are effective in enhancing electrochemical performance by increasing the specific surface area and minimizing ion diffusion pathways, the concurrent realization of uniform particle morphology, narrow size distribution, high crystallinity, and phase purity remains a formidable challenge. Synthesis conditions that favor high crystallinity, involving elevated temperatures and extended reaction durations, tend to promote excessive particle growth and agglomeration, thereby diminishing the advantages conferred by nanoscale structuring. Conversely, conditions conducive to the formation of fine nanostructures often result in reduced crystallinity or the emergence of amorphous phases, which can compromise the structural stability of the material during prolonged electrochemical cycling.

The scalability and cost-efficiency of hydrothermal and solvothermal synthesis routes for high-entropy oxides are pivotal factors for their practical application, yet they present considerable engineering and economic obstacles. Laboratory-scale methodologies often face significant limitations when transitioning to industrial-scale production. The need for sealed, high-pressure autoclaves introduces complex challenges related to reactor design, operational safety, and maintenance, challenges that are further amplified in large-scale systems. Achieving uniform heat and mass transfer within large reactors to ensure consistent product quality remains a critical engineering barrier. Additionally, the extended reaction durations typically required to attain adequate crystallinity and phase purity at relatively low hydrothermal temperatures constrain production throughput and elevate energy consumption per unit of output [114]. Furthermore, the high cumulative cost associated with multiple high-purity metal precursors, combined with solvent demands and waste management considerations, can substantially undermine the economic viability of these synthesis methods relative to more established anode material production processes, unless systematically optimized [115].

Reproducibility remains a fundamental challenge in the hydrothermal and solvothermal synthesis of complex materials, such as high-entropy oxides. The physicochemical properties of the resulting HEOs are highly sensitive to subtle fluctuations in numerous synthesis parameters, including precursor concentrations, solution pH, internal temperature gradients within the autoclave, agitation rates, and the purity of reagents. In multi-component systems, the interplay among these variables is often intricate and non-linear, such that minor deviations can result in pronounced batch-to-batch inconsistencies in phase composition, particle size distribution, and morphological characteristics [116]. Developing robust and reproducible synthesis protocols thus necessitates a comprehensive understanding of these sensitivities, along with meticulous process control, are requirements that are difficult to fulfill without extensive empirical optimization and precise instrumentation.

4.5. *Spray Pyrolysis of High-Entropy Oxides*

The synthesis of high-entropy oxides via spray pyrolysis represents a promising strategy for the development of compositionally complex materials as anode candidates in lithium-ion batteries, potentially offering improved electrochemical performance. Nonetheless, this approach is accompanied by several critical challenges, both methodological and intrinsic to the nature of HEOs, which must be systematically addressed to optimize the structural and functional properties of the resulting materials.

Attaining accurate stoichiometric control and uniform elemental dispersion within multi-component oxides remains a significant challenge. Variations in the solubility, thermal

decomposition behavior, and reaction kinetics of different metal precursors within the sprayed droplets often result in compositional inhomogeneities and phase segregation, thereby impeding the formation of a genuine single-phase solid solution [117].

Regulating particle morphology, size distribution, and crystallinity represents an additional major challenge. Although spray pyrolysis typically produces spherical particles, achieving a narrow particle size distribution, appropriate porosity for effective electrolyte infiltration and Li-ion transport, and adequate crystallinity directly from the synthesis process remains difficult. Frequently, the as-prepared particles are amorphous or exhibit low crystallinity, necessitating post-synthesis annealing. However, this additional thermal treatment can promote undesirable grain growth, compromise the intended nanostructure, and increase both processing complexity and overall production costs.

Moreover, achieving phase purity is essential. The inherently rapid heating and cooling rates associated with spray pyrolysis may be insufficient to facilitate complete elemental diffusion and incorporation into a single-phase high-entropy oxide lattice [118,119]. Consequently, this can lead to the emergence of secondary phases or residual unreacted metal oxides, both of which adversely affect the electrochemical stability and capacity of the anode material.

The selection of suitable precursors and the fine-tuning of solution chemistry pose additional significant challenges. The precursor solution must exhibit chemical stability, preventing premature precipitation of individual constituents, while the decomposition mechanisms of the mixed precursors must be mutually compatible to enable the formation of the targeted high-entropy oxide phase [119].

Although spray pyrolysis is frequently highlighted for its scalability, achieving consistent quality and high yields of high-entropy oxide powders with the desired physicochemical properties during large-scale production necessitates precise control over a multitude of processing parameters, including precursor solution feed rate, carrier gas flow rate, reactor temperature profile, and droplet size distribution. Addressing these challenges is essential for unlocking the full potential of spray-pyrolyzed HEOs as high-performance anode materials for next-generation lithium-ion batteries [120].

4.6. Chemical Vapor Deposition of High-Entropy Oxides

Chemical Vapor Deposition (CVD) represents a highly adaptable thin-film deposition method, capable of yielding high-quality, conformal high-entropy oxide coatings. Nevertheless, several inherent challenges currently hinder the broad implementation of CVD for the fabrication of HEO-based anodes in lithium-ion batteries. Overcoming these limitations is essential for fully leveraging the advantages of HEOs in the development of next-generation energy storage technologies.

One of the main challenges associated with using CVD in obtaining HEOs for LIB anodes is the compositional control and uniformity. Achieving and maintaining the desired stoichiometry across a large substrate during CVD can be challenging due to varying precursor volatilities, decomposition temperatures, and reaction kinetics for each element [121]. This can lead to elemental segregation or the formation of undesired secondary phases. Precursor selection for a wide range of metallic elements that are compatible with the CVD process can be difficult, along with maintaining a uniform composition across the thickness of the deposited film. Gradients in elemental distribution can impact the electrochemical performance.

The high entropy effect aims to stabilize a single-phase solid solution. However, achieving this phase purity via CVD can be difficult, as competitive phases might form depending on deposition parameters (temperature, pressure, precursor flow rates, substrate). Impurities or secondary phases can negatively affect cycling stability and capacity.

Controlling the crystallinity (amorphous vs. crystalline) and grain size is crucial for electrochemical performance. CVD parameters need to be precisely tuned to obtain the desired grain size, which impacts surface area for Li diffusion and volume expansion accommodation. For anode materials, a porous structure and high surface area are generally beneficial for enhancing Li^+ diffusion kinetics and accommodating volume changes during lithiation/delithiation. Ensuring good adhesion of the HEO film to the current collector is critical to prevent delamination during repeated cycling due to volume changes [122]. The mechanical stability of the deposited film under electrochemical stress is also a concern.

CVD of HEOs requires the simultaneous delivery and controlled mixing of multiple precursor gases. This adds significant complexity to the reactor design, gas manifold system, and process control compared to single or binary compound deposition. Achieving and maintaining uniform temperature across a large substrate area in a CVD reactor is crucial for consistent film quality and composition. While CVD offers excellent control, it can be a relatively expensive process compared to some wet chemistry routes, especially for large-scale production. Optimizing the process for high throughput and lower cost is essential for commercial viability. Another important aspect is that some CVD precursors can be toxic, flammable, or corrosive, necessitating strict safety protocols and waste management.

HEOs, especially transition metal oxides, can react with the electrolyte to form a solid-electrolyte interphase (SEI) layer. The nature and stability of this SEI layer greatly influence the battery life cycle and coulombic efficiency. CVD parameters can influence the surface chemistry of the HEO, which in turn affects SEI formation.

Although HEOs generally show improved structural stability compared to single-component metal oxides, they still undergo volume changes during the lithiation/delithiation process. Designing CVD parameters to create robust HEO structures that can tolerate these changes is crucial to prevent pulverization and capacity fade [122,123].

Addressing these challenges requires a multidisciplinary approach involving advanced material synthesis, in-depth characterization techniques, and computational modeling to understand the complex interplay between obtaining process parameters, material properties, and electrochemical behavior.

4.7. Interface Compatibility of Anode Materials with Sulfide and Oxide Solid Electrolytes in All-Solid-State Batteries

Beyond conventional cells based on liquid electrolytes, all-solid-state batteries (ASSBs) employing sulfide (e.g., $\text{Li}_6\text{PS}_5\text{Cl}$, $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$) or oxide (e.g., garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, LLZO) solid electrolytes are increasingly regarded as the next step in the evolution of Li-ion technology, owing to their improved safety and potentially higher energy density [124]. For the anode materials reviewed in this work, the transition from liquid to solid electrolytes replaces the classical SEI problem with a solid–solid interface problem, in which the chemical and electrochemical compatibility with the electrolyte, the interfacial resistance and the mechanical integrity of the contact become the dominant performance-limiting factors [124,125].

Sulfide solid electrolytes offer high ionic conductivities and good ductility, which facilitates intimate contact with the anode by simple cold pressing. However, their electrochemical stability window is narrow, and they are reduced at the low operating potentials of graphite, silicon or conversion-type oxide anodes, forming decomposition products (e.g., Li_2S , Li_3P , LiCl) that increase the interfacial impedance [125]. For silicon, this interfacial reactivity can even be turned into an advantage: Tan et al. [126] demonstrated that the passivating interphase formed between micro-silicon and a $\text{Li}_6\text{PS}_5\text{Cl}$ electrolyte suppresses the continuous interfacial growth and the irreversible lithium losses typical of liquid cells, enabling carbon-free, high-loading (99.9 wt.%) micro-silicon anodes with stable full-cell cycling. For conversion-type oxides, including HEOs, the large volume changes during

cycling additionally cause contact loss and void formation at the rigid solid–solid interface, so the applied stack pressure and the electrode architecture must be engineered together with the material itself [124,125].

Oxide solid electrolytes, such as garnet-type LLZO, are chemically more robust and possess wider electrochemical stability windows, but their rigidity leads to poor physical contact and very high interfacial resistance with the electrode materials. Surface engineering has proven to be an efficient modification strategy in this case: Han et al. [127] showed that an ultrathin Al_2O_3 layer deposited by atomic layer deposition (ALD) on the garnet surface reduces the interfacial impedance from $1710\ \Omega\cdot\text{cm}^2$ to about $1\ \Omega\cdot\text{cm}^2$, by improving the wettability and enabling efficient lithium-ion transport across the interface.

Consequently, the modification strategies discussed in the previous sections for liquid-electrolyte systems—conformal carbon coatings, artificial buffer layers (such as ALD-deposited Al_2O_3), nanostructuring to accommodate volume changes and composite electrode designs in which the active material is intimately mixed with the solid electrolyte—constitute the main toolbox for matching the reviewed anodes with sulfide and oxide solid electrolytes [124–127]. Among the reviewed materials, LTO is particularly well suited to solid-state operation due to its zero-strain behavior and to its higher operating potential, which lies within the stability window of most solid electrolytes, whereas the compatibility of HEO anodes with solid electrolytes remains essentially unexplored and represents an important direction for future research.

4.8. Industrial Bottlenecks and Optimization Approaches for the Commercialization of HEO Anodes

Although HEO anodes have been emphasized throughout this review for their promising electrochemical behavior, their transfer from the laboratory to the industrial scale is currently limited by several bottlenecks. From an economic perspective, HEOs require five or more high-purity metal precursors, some of which (particularly cobalt and nickel) are costly, supply-critical and environmentally problematic, so the raw-material cost per kilogram of active material is substantially higher than for graphite [128,129]. From a manufacturing perspective, maintaining the strict near-equimolar stoichiometry and the single-phase structure over large production batches is challenging, since minor deviations in composition or thermal history can lead to secondary phases and pronounced batch-to-batch variability, as discussed in Sections 4.3–4.6 for the individual synthesis routes [129,130].

From an electrochemical perspective, conversion-type HEO anodes suffer from a considerable irreversible capacity loss in the first cycle, from voltage hysteresis between lithiation and delithiation, and from an average working potential that penalizes the energy density at the full-cell level [128,130]. In addition, the nanostructured powders, which give HEOs their attractive rate performance, also exhibit low tap density, translating into a low volumetric energy density at the electrode level, and their compatibility with industrial slurry processing, binders and calendaring has not yet been systematically demonstrated. Finally, most reported results are obtained in half cells over a limited number of cycles, and standardized full-cell validation under industrially relevant conditions is still lacking [130,131].

Several optimization approaches are being pursued to address these limitations. Compositional design increasingly targets the replacement of critical elements with abundant ones (Fe, Mn, Zn, Mg) while preserving the entropy stabilization; carbon composites and conductive coatings are used to improve the electronic conductivity and the first-cycle efficiency; prelithiation strategies can compensate for the initial irreversible lithium loss; and scalable synthesis routes with good stoichiometric control—co-precipitation, spray pyrolysis, solution combustion and the induction melting/oxidative milling hybrid route

described in Section 3.7.2, which additionally allows the use of recycled feedstocks—are being adapted to HEO production [128–131]. In parallel, machine-learning-assisted screening of the vast HEO compositional space is expected to accelerate the identification of compositions that combine electrochemical performance with low cost and industrial processability [130,131].

5. Methods for Reducing the Environmental Impact Through the Usage of Unconventional Obtaining Methods

High-entropy oxides are promising anode materials for lithium-ion batteries due to their high energy density, excellent stability, and versatile compositions. However, conventional synthesis methods for HEOs can be energy-intensive and rely on potentially environmentally impactful precursors. To reduce the environmental footprint, researchers are exploring various unconventional and green synthesis methods. Among the methods of reducing the environmental impact through the usage of unconventional techniques are the replacement of hazardous precursors, opting for metal salts with lower toxicity, or using water as a solvent instead of organic solvents to reduce the volatile organic compound emissions or solvent wastes.

Another important aspect is exploring and integrating mixed metal scraps, metallic elements resulting from battery waste processing, as raw material sources for high entropy oxide synthesis. Applying these measures has the ability to reduce the demand for raw materials and minimize waste generation after the end of the Li-ion batteries' life duration. This aspect has a long-term favorable impact on the environment and can bring significant improvements in the safe-and-sustainable-by-design concept for new materials by facilitating its practical implementation and encouraging recyclability.

Prioritizing the incorporation of abundant and less critical elements in HEO compositions can reduce reliance on expensive and environmentally problematic elements like cobalt and nickel, contributing to both economic and environmental sustainability.

In the meantime, another method could be increasing the efficiency of the synthesis process by using a microwave field, since it can significantly reduce the reaction time and energy consumption, compared to other conventional heating methods. On the other hand, an indispensable component in the synthesis process, independent to the selected method, is utilizing the computational tools, including artificial intelligence means, to predict optimal compositions and processing routes to reduce the need for extensive experimental trials, saving resources, time and energy in materials design and optimization. Such data-driven screening approaches, combined with the experimental synthesis routes discussed in this review, form the basis of a rational, model-guided design of anode materials. Taken together, these measures embody the safe-and-sustainable-by-design (SSbD) framework, in which safety, environmental impact and circularity criteria are integrated into the material development process from the earliest design stages, rather than being assessed retrospectively.

6. Conclusions

This review has presented a comprehensive overview of current anodic materials and synthesis strategies for lithium-ion batteries, highlighting both the remarkable progress and the persistent challenges in the field. Traditional materials like graphite remain dominant due to their stability and cost-effectiveness, but emerging alternatives such as silicon, lithium titanate (LTO), tin-based compounds, carbon nanostructures (CNTs and graphene) and high entropy oxides (HEOs) provide new ways to improve capacities, rate capabilities and electrochemical performance.

The synthesis methods examined—from conventional solid-state reactions to advanced techniques like electrospinning and hybrid approaches—demonstrate the decisive influence of the processing route on the morphology, phase purity, surface properties and, ultimately, the electrochemical performance of anode materials, while computational tools are emerging as valuable aids for guiding composition and process optimization. Furthermore, the environmental and economic aspects of material processing are gaining increasing attention, with unconventional and green synthesis routes offering promising avenues to reduce ecological impact and enhance sustainability.

Despite these advancements, major challenges continue to be problematic, including structural degradation during cycling, unstable solid–electrolyte interphase (SEI) formation, high production costs and scalability concerns. The successful development of next-generation anodes depends on the ability to engineer safe and sustainable-by-design materials with controlled nanostructures, stable interfaces and environmentally benign production methods. Equally important, the interfacial compatibility of these anodes with sulfide and oxide solid electrolytes, as well as the industrial bottlenecks that currently limit the commercialization of emerging materials such as HEOs, both discussed in this review, will have to be addressed before next-generation battery configurations can reach the market.

Future research should continue to focus on innovative material design, scalable and sustainable synthesis techniques and in-depth electrochemical characterization. These combined efforts will be crucial for enabling the commercial viability of advanced anode materials and for supporting the continued evolution of lithium-ion battery technologies to keep up the demands of portable electronics, electric vehicles and grid-scale energy storage.

Author Contributions: Conceptualization, B.-A.Ş., I.-C.B. and Ş.C.; methodology, Ş.C. and D.-F.M.; software, B.F., D.M. and I.A.; validation, M.-T.O. and M.C.; formal analysis, M.C. and L.M.C.; investigation, M.-T.O. and S.-A.F.; resources, I.-C.B. and L.M.C.; data curation, B.F. and I.A.; writing—original draft preparation, I.-C.B., Ş.C. and D.M.; writing—review and editing, B.-A.Ş. and D.-F.M.; visualization, M.-T.O. and S.-A.F.; supervision, M.C.; project administration, B.-A.Ş.; funding acquisition, B.F. and D.-F.M. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Core Program within the National Research Development and Innovation Plan 2022–2027, financed by MCID, project no. PN 23 25 01 03 and by a grant of the Ministry of Research, Innovation and Digitization, CNCS/CCCDI-UEFISCDI, project number ERANET-ERAMIN-3-HETMOLION, within PNCDI IV. The APC was funded by University Politehnica of Bucharest, Romania, within the PubArt Program.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding authors.

Conflicts of Interest: The authors declare no conflicts of interest.

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